

# Smaran Vallabhaneni

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## EDUCATION

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| <b>Olin College of Engineering</b><br><i>Bachelor of Science, Engineering in Computing (GPA: 4.0)</i>                                                                                                                                                                                                                                                                                                                           | Needham, MA<br><i>Aug. 2025 – May 2029</i> |
| – <b>Relevant Courses:</b> Linear Algebra, Multi-Variable Calculus, Modeling & Simulation, Bayesian Statistics, Machine Learning, Software Design (Python), Kernel Development (C), Algorithms and Data Structures, Cryptography, Data Science, Statistics for Data Science, Data Analysis with Python, Data Preparation & Applied Machine Learning, Products & Markets, Introduction to Sensors, Instrumentation & Measurement |                                            |

## SKILLS

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| <b>Programming Languages:</b> Java, C, C++, C#, R, Python, Rust, NumPy, pandas, MySQL, NoSQL, Swift, HTML, CSS, JavaScript, TypeScript, Dart, L <sup>A</sup> T <sub>E</sub> X, Bash, MATLAB                           |  |
| <b>Operating Systems:</b> Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, Fedora, and Arch based)                                                                                                               |  |
| <b>Tools:</b> GitHub Copilot, Google Gemini, Google Antigravity, Claude Code, LM Studio, Git, IntelliJ, Eclipse IDE, VSCode, Xcode, Node.js, Next.js, Tableau Desktop, Docker, ArduPilot, Figma, Adobe Creative Suite |  |
| <b>Frameworks:</b> LLMs, RAG Systems, Ollama, PyTorch, Scikit-Learn, OpenCV, MediaPipe, Django, AngularJS, React, Kivy, Flutter, SwiftUI, ROS, MAVROS                                                                 |  |
| <b>Other:</b> HuggingFace, NLP, Amazon Web Services, Google Cloud, Microsoft Azure, Dev Ops, API Integration, Testing and Debugging, Continuous Integration Pipelines, UI, UX                                         |  |
| <b>Soft Skills:</b> Communication, Problem Solving, Team Collaboration, Leadership, Adaptability, Time Management, Attention to Detail                                                                                |  |

## PROJECTS

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>W.A.G.E</b>   <i>Python, TypeScript, Next.js, React, SQL</i>                                                                                                                                                                                               | Jul. 2026             |
| – Built a local-first AI job search strategy tool designed to make pre-application strategies more streamlined, with a fit score, tailored resume bullets, and outreach drafting, integrating chatgpt-oss-20b and nomic embeddings for high accuracy parsing. |                       |
| – Implemented robust SQL database and REST API principles with optional Ollama Cloud integration, with response times running 5 seconds for resume to results workflow under intensive 25+ job descriptions at a time.                                        |                       |
| <b>Specter</b>   <i>Python, Linux, LLMs</i>                                                                                                                                                                                                                   | May 2026              |
| – Created a local-first, privacy oriented AI screen agent for Fedora Wayland to perform tasks efficiently, leveraging gemma4 vision models to parse content at a response time of 3 seconds per action, with an 86% task success rate.                        |                       |
| <b>3D Gesture Tracking</b>   <i>Blender, Python, OpenCV</i>                                                                                                                                                                                                   | Apr. 2026             |
| – Engineered a real-time computer vision add-on for Blender, achieving a sub-50 ms response time and a low 5% false-positive rate across diverse user testing profiles.                                                                                       |                       |
| <b>Whoa-Scope</b>   <i>Python, Kivy</i>                                                                                                                                                                                                                       | Feb. 2026 – Apr. 2026 |
| – Co-maintained an open-source oscilloscope software fork to resolve compatibility issues and desktop UX flaws, gaining organic adoption from 50% of the class (40 students) and faculty recognition.                                                         |                       |
| – Profiled and refactored memory allocation, slashing RAM consumption by 200+ MB to maximize stability and prevent lag on lab computers.                                                                                                                      |                       |
| <b>ClassicTunes</b>   <i>Swift, SwiftUI, Metal API</i>                                                                                                                                                                                                        | Oct. 2025 – Jul. 2026 |
| – Developed an open-source modern remake of iTunes 7–10 for Apple Silicon macOS, faithfully recreating CoverFlow, MiniPlayer, the iTunes Store, iPod Syncing, and Smart Playlists.                                                                            |                       |
| – Leveraged Apple’s Metal API to build a custom GPU-accelerated CoverFlow engine, running with smaller memory and CPU footprint of the native Apple Music client with a fluid 120 FPS render rate across 150+ albums on Apple Silicon.                        |                       |

## EXPERIENCE

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| <b>CLOS (Command Line Operating System)</b><br><i>Founder and Lead Engineer</i>                                                                                                                                                                              | Remote<br><i>Apr. 2023 – Present</i>      |
| – Building an offline-first software providing accessible mathematical and scientific computing for students, educators, and professionals.                                                                                                                  |                                           |
| – Engineered a custom database engine, symbolic math parser, custom scripting language, and a 3D rendering library from scratch in Java; achieved consistent 60 FPS graphic rendering and executed complex multi-variable calculus computations under 50 ms. |                                           |
| <b>Olin College AERO Project Team</b><br><i>Head of Ground Station Development — Software Engineer</i>                                                                                                                                                       | Needham, MA<br><i>Sep. 2025 – Present</i> |
| – Leading software development for an autonomous fixed-wing aircraft competing in the AUVSI SUAS competition, targeting precision water delivery to GPS-specified targets using image recognition.                                                           |                                           |
| – Independently architected ground station software stack integrating ArduPilot, MissionPlanner, ROS, and MAVROS systems to bridge an on-board NVIDIA Jetson computer with ground controls.                                                                  |                                           |
| – Engineered a robust fly-by-wire telemetry pipeline, achieving a consistent sub-100ms round-trip latency for real-time sensor monitoring and safety interlocks.                                                                                             |                                           |
| <b>Geeky Insights</b><br><i>Full-Stack Android App Developer — Internship</i>                                                                                                                                                                                | Remote<br><i>Apr. 2024 – Apr. 2025</i>    |
| – Designed end-to-end UI and UX of a customizable audio alarm clock app using Figma and Adobe Creative Suite, prioritizing accessibility and intuitive design for a diverse user base.                                                                       |                                           |
| – Engineered an end-to-end front-end and backend architecture using Flutter, Dart, and Java, optimizing the compilation pipeline into a lightweight, bug-free 15 MB production package.                                                                      |                                           |
| – Optimized application memory layout to support low-resource devices running Android 9+, reducing RAM consumption to match the baseline footprint of Android’s native system clock.                                                                         |                                           |

## CERTIFICATIONS

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|------------------------------------------------------------|-------------------------------------------------|
| <b>Meta Database Engineer Certification</b><br><i>Meta</i> | Jul. 2025<br><i>Credential ID: M8WKYG9PMT4G</i> |
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